

Self-improving generative foundation model for synthetic medical image generation and clinical applications

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In many clinical and research settings, the scarcity of high-quality medical imaging datasets has hampered the potential of artificial intelligence (AI) clinical applications. This issue is particularly pronounced in less common conditions, underrepresented populations and emerging imaging modalities, where the availability of diverse and comprehensive datasets is often inadequate. To address this challenge, we introduce a unified medical image–text generative model called MINIM that is capable of synthesizing medical images of various organs across various imaging modalities based on textual instructions. Clinician evaluations and rigorous objective measurements validate the high quality of MINIM’s synthetic images. MINIM exhibits an enhanced generative capability when presented with previously unseen data domains, demonstrating its potential as a generalist medical AI (GMAI). Our findings show that MINIM’s synthetic images effectively augment existing datasets, boosting performance across multiple medical applications such as diagnostics, report generation and self-supervised learning. On average, MINIM enhances performance by 12% for ophthalmic, 15% for chest, 13% for brain and 17% for breast-related tasks. Furthermore, we demonstrate MINIM’s potential clinical utility in the accurate prediction of HER2-positive breast cancer from MRI images. Using a large retrospective simulation analysis, we demonstrate MINIM’s clinical potential by accurately identifying targeted therapy-sensitive *EGFR* mutations using lung cancer computed tomography images, which could potentially lead to improved 5-year survival rates. Although these results are promising, further validation and refinement in more diverse and prospective settings would greatly enhance the model’s generalizability and robustness.

Availability of large healthcare datasets is pivotal for driving artificial intelligence (AI) model development and clinical applications. However, privacy concerns pose important ethical and legal issues, making it difficult to share such data. The scarcity of high-quality imaging datasets has hindered the integration of cutting-edge AI technologies into medicine and healthcare applications. To address this challenge, data augmentation and the use of synthetic images with generative AI have emerged as key solutions. Generative AI has enabled major

breakthroughs in medical imaging research, enhancing the analysis of medical records and images and improving diagnostics and treatment planning^{1–4}. Recent efforts have demonstrated that generative AI can synthesize high-quality chest X-ray images^{5–7}, morphologically preserving three-dimensional (3D) images of the brain⁸ and two-dimensional (2D) images of histopathology^{9,10} and dermatology^{10,11}, and consequently enhancing medical image understanding and improving downstream analysis.

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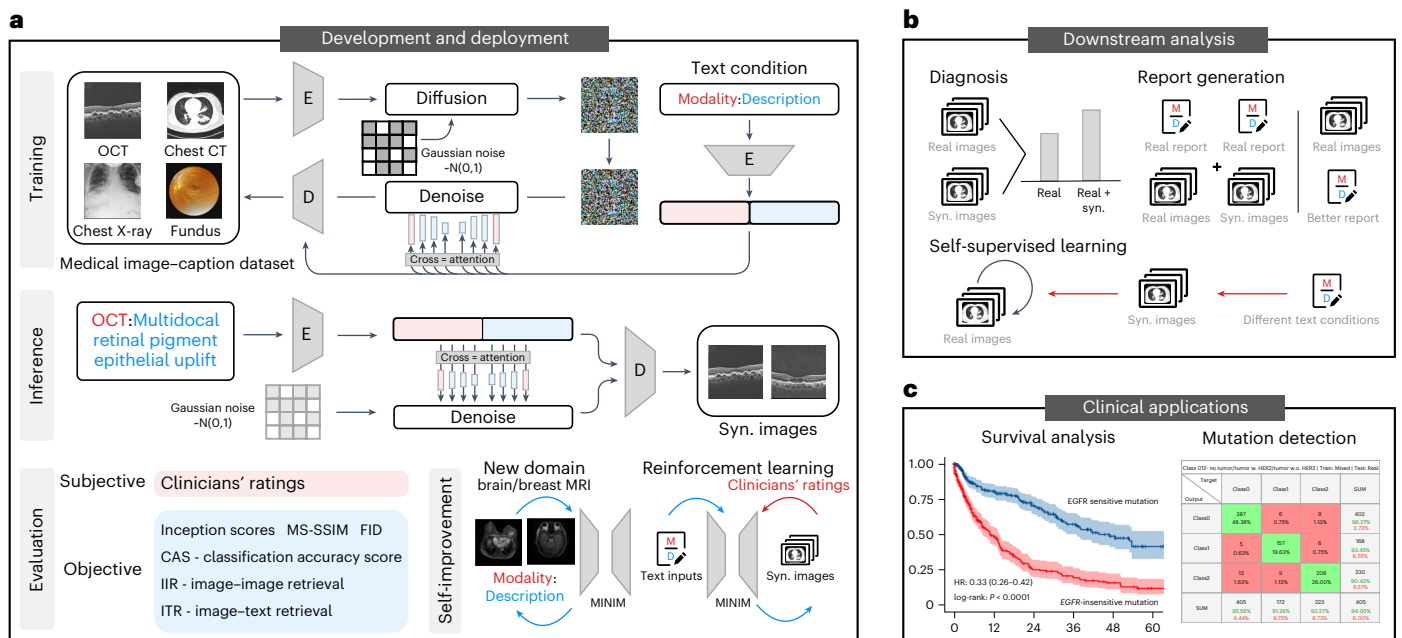


Fig. 1 | Schematic illustration of our generative system for medical image synthesis. a, MINIM's development and deployment. The training was conducted with a latent stable diffusion model on paired images and reports from different modalities and organs. In deployment, MINIM is able to generate high-quality synthetic images given diverse textual descriptions. The synthetic images were evaluated through subjective assessments from clinicians and multiple objective metrics. **b**, The synthetic images can be used as an additional

training source and are demonstrated in three scenarios, including diagnosis, report generation and self-supervised learning. Icons are from references <https://uxwing.com/image-icon/> and <https://uxwing.com/edit-document-icon/>. **c**, The synthetic images can assist clinical applications, including accurate prediction of mutations of HER2 in breast cancer MR images and EGFR in lung cancer CT images, and improve survival analysis. Syn., synthetic.

Generative adversarial networks (GANs), in particular, have been extensively employed for generating synthetic medical images, showing remarkable progress in enhancing data availability for AI models^{12,13}. Additionally, newer pipelines built upon GANs have also incorporated segmentation techniques to further reduce the required amount of dataset of training, which is tailored for data-scarce scenarios, such as in cases of rare diseases¹⁴. These studies have shown that, within a single medical imaging modality, generative AI can generate high-quality synthetic images that act as valuable assets to support medical imaging research.

However, traditional GAN-based image generation struggles with generating images of diverse dimensionalities and is, thus, often limited to a single imaging modality¹⁵. The limited exploration of one modality and the lack of investigation into the relationships among different medical imaging modalities further hinder the full utilization of extensive multimodal medical image datasets and the development of universal medical generative models.

Here we introduce a unified medical image-text generative model called MINIM. MINIM integrates medical images paired with textual descriptions across various modalities and organs, including optical coherence tomography (OCT), fundus, chest X-ray and chest computed tomography (CT). Our research demonstrated MINIM's capability to generate high-quality synthetic images for each organ and imaging modality based on textual descriptions. We evaluated MINIM's performance in generating synthetic images based on textual inputs, and its performance was tested across different medical imaging scenarios against other recent generative models^{16–19}. We also explored the model's adaptability by incorporating additional magnetic resonance imaging (MRI) datasets for brain and breast imaging, allowing us to assess MINIM's potential for continuous learning and integration of new medical knowledge.

Results

MINIM's framework overview

To develop a versatile and generalizable medical image-text generative model, we used a range of medical imaging modalities, including OCT, fundus, chest X-ray and chest CT, paired with corresponding textual descriptions as training data (Methods). Our model, MINIM, operates through two major phases: an initial developmental phase that integrates all medical images and text descriptions into a stable diffusion model and a deployment phase, where synthetic images are generated based on textual inputs (Fig. 1a). To comprehensively assess MINIM's capabilities, we conducted a multi-faceted evaluation: (1) the quality of synthetic images was evaluated using both objective metrics and subjective assessments by clinicians; (2) we explored downstream applications of the synthetic images in diagnostics, report generation and self-supervised learning (Fig. 1b); (3) we implemented self-improvement strategies for MINIM with reinforcement learning on human feedback (RLHF) and transfer learning (Fig. 1a); and (4) we applied MINIM to clinical tasks in detecting mutation and survival analysis (Fig. 1c).

Quality evaluations

To test the quality of synthetic images generated by MINIM, we conducted a comprehensive set of assessments, including both subjective and objective assessments. We began with subjective evaluations by clinicians followed by objective metrics, including Fréchet inception distance (FID)²⁰, inception score (IS)²¹ and multi-scale structural similarity index measure (MS-SSIM)²², to measure fidelity, diversity and quality. Additionally, the classification accuracy score (CAS), zero-shot image-image retrieval (IIR) and image-text retrieval (ITR) methods were incorporated into the assessment to assess the practical utility and relevancy of synthetic images. All evaluations were performed on OCT, fundus, chest X-ray and chest CT. Training details and comparative image-text generative models can be found in the Methods.

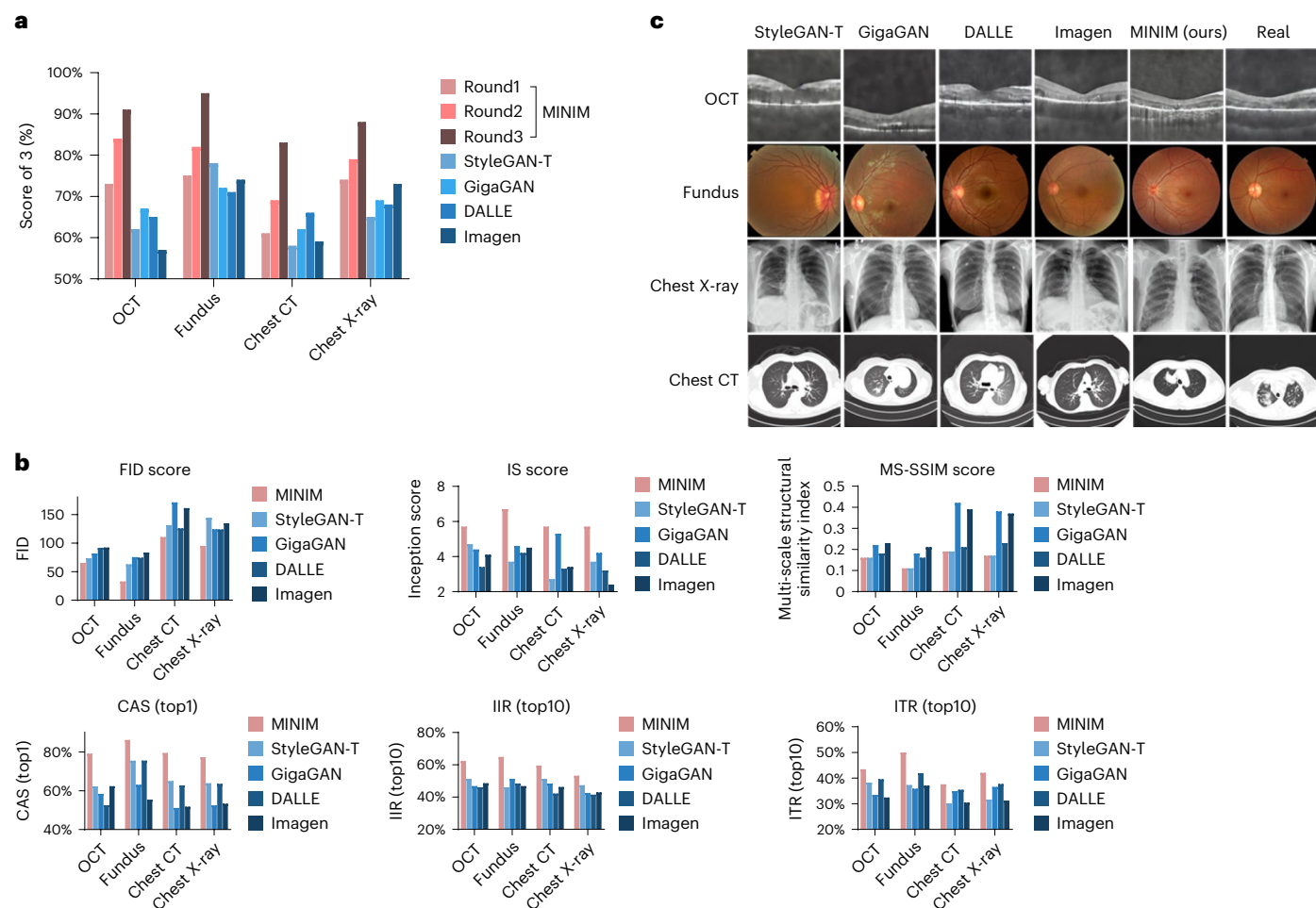


Fig. 2 | Evaluations of generated synthetic images. a, Clinicians conducted three rounds of scoring on the generated synthetic images. A score of 1 denoted a low-quality image; a score of 2 indicated a high-quality image but one that was irrelevant to the report; and a score of 3 signified a high-quality image that aligned with the report. Based on the scoring results, we trained the two-stage RL

strategy model and iteratively applied it to MINIM. **b**, Performance comparison of objective assessments from six objective metrics (FID, IS, MS-SSIM, CAS, IIR and ITR) using MINIM and other text-to-image generative models. **c**, Demonstration of the synthetic images using different methods.

Subjective assessment from clinicians

We invited clinicians specific for corresponding image modalities to assess whether synthesized images are clinically accurate and useful (Fig. 2a). Clinicians rated the synthetic images on a scale of 1 to 3: a score of 1 denoted a low-quality image; 2 indicated a high-quality image but irrelevant to the report; and 3 signified a high-quality image that aligned well with the report. We conducted three rounds of ratings and summarized the results by calculating the percentage of images that scored 3. In the first round, MINIM achieved an average of 70.75% (73% for OCT, 75% for fundus, 61% for chest CT and 74% for chest X-ray), whereas the best competing generative models (StyleGAN-T) achieved an average of 65.75% (62% for OCT, 78% for fundus, 58% for chest CT and 65% for chest X-ray). In the third round, after implementing reinforcement learning (RL) (Methods), MINIM's performance showed substantial improvement, achieving an average of 89.25% (91% for OCT, 95% for fundus, 83% for chest CT and 88% for chest X-ray).

Objective image quality assessment

We employed objective metrics to quantitatively assess the fidelity, diversity and overall quality of the generated images. These images should show robust diversity to encompass a wide range of pathological variations without overfitting to the training data while also maintaining high fidelity to real images, as reflected in the low FID, low

MS-SSIM and high IS scores (Extended Data Fig. 1). FID was calculated for each synthetic–real image pair, whereas IS and MS-SSIM were computed independently among synthetic images. We reported the average results of MINIM and other competitive text-to-image generative methods (Fig. 2b). Our findings revealed that MINIM's performance (FID: 65.3, IS: 5.7 ± 0.42 , MS-SSIM: 0.16 ± 0.03 for OCT; FID: 32.7, IS: 6.7 ± 0.43 , MS-SSIM: 0.11 ± 0.03 for fundus; FID: 110.4, IS: 5.7 ± 0.37 , MS-SSIM: 0.19 ± 0.06 for chest X-ray; FID: 94.8, IS: 5.7 ± 0.38 , MS-SSIM: 0.17 ± 0.04 for chest X-ray) outperformed other generative methods by a large margin. Samples of MINIM's synthetic images along with other generative methods indicate that MINIM can produce more realistic images in all settings (Fig. 2c).

CAS performance

The CAS is an emerging proxy for the evaluation of synthetic data²³. CAS measures how accurately models trained exclusively on synthetic samples can classify real test data. For each of the four modalities and organs, we used MINIM to generate synthetic images 10 times the real dataset for each class label (Methods) and reported CAS on the test set using 1/10 random images of each real dataset. Figure 2b shows that MINIM outperformed other generative methods for top-1 accuracy across different datasets (79.09% for OCT, 86.16% for fundus, 79.42% for chest CT and 77.23% for chest X-ray).

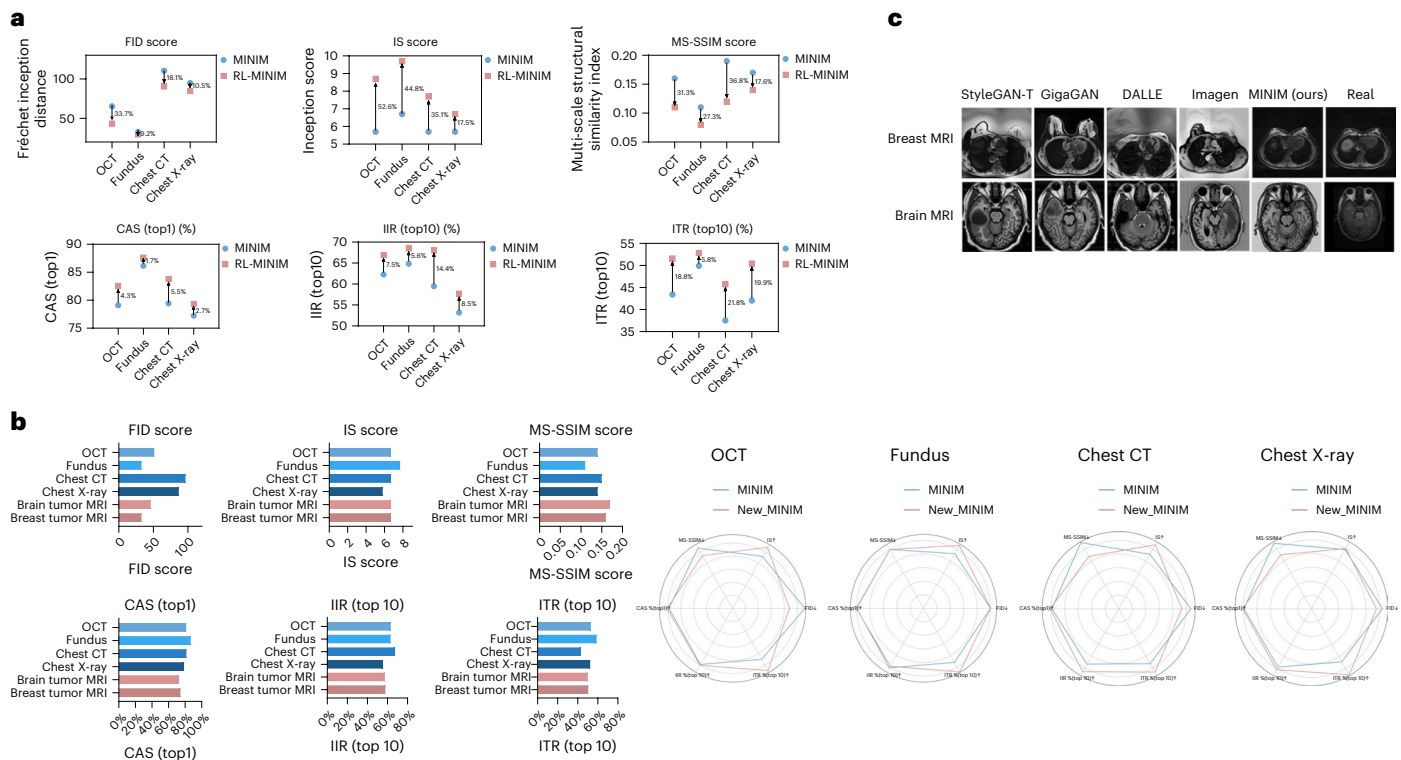


Fig. 3 | Self-improvement evaluations. **a**, Six subjective results of the synthetic images using RLFH from clinicians' ratings. **b**, Performance improvement of MINIM's generative ability by accommodating image-text paired data from new domains. **c**, Demonstration of the synthetic images in new domains using MINIM and other generative methods.

Performance on zero-shot IIR

To further validate the quality of the synthetic images from MINIM, we employed a zero-shot IIR method. Using real images as queries, this evaluation aimed to retrieve candidate synthetic images that are closely related in category to the query image. Both the query and candidate images were input into a pre-trained image encoder, specific to their imaging modality, to rank the candidate images based on feature similarity to the query image using cosine similarities. We reported the number of synthetic images that share the same category as queried real images among the top 10 candidates, using precision@k metrics where $k = 10$, for each query. Figure 2b demonstrated the superior results (62.25% for OCT, 64.83% for fundus, 59.44% for chest CT and 53.17% for chest X-ray) from MINIM to other generative models in each case.

Performance on zero-shot ITR

To evaluate whether synthesized images correspond to clinically relevant and accurate texts, we performed zero-shot ITR. In this approach, a query image embedding is mapped into a textual embedding space to retrieve the most relevant textual impression corresponding to the image. We calculated retrieval precision using the precision@k with $k = 10$, where a point of accuracy is awarded for each textual impression that shares the same category as the query image. Figure 2b demonstrated the superior results (43.41% for OCT, 49.93% for fundus, 37.53% for chest CT and 42.04% for chest X-ray) from MINIM compared to other generative models.

Self-improving strategies

RL from clinicians' ratings. The RLHF strategy uses a synthetic-in-the-loop approach to iteratively enhance MINIM's performance, which involves leveraging clinicians' ratings, or feedback, to create a closed loop of continuous learning. Our two-stage RLHF strategy began with a panel of clinicians who assessed the quality of synthetic images in relation to their corresponding prompt text reports. Clinician feedback

was then used to train a reward model that mimics the clinicians' ratings. This reward model was incorporated into MINIM, serving as an additional layer of supervision to guide the model in training and improve overall performance. We compared how RLHF incorporation improved image generation compared to four other data augmentation methods across six subjective metrics (Fig. 3a). Notably, all five methods, including MINIM, exhibited improvements over their original generative models, with the two-stage RLHF strategy showcasing the most significant enhancements (for example, for OCT, FID down to 43.3 versus 65.3; IS up to 8.7 ± 0.37 versus 5.7 ± 0.42 ; and MS-SSIM down to 0.11 ± 0.06 versus 0.16 ± 0.03). These results underscore the value of integrating human feedback into the RL process, substantially refining the generative model's performance across multiple metrics.

Transfer learning. A major advantage of MINIM over previous specialized models is its ability to leverage image-text paired medical data from various organs and imaging modalities. After the inclusion of breast and brain MRI datasets in a new round of development, we evaluated the performance of previously trained modalities (Fig. 3b, right panel). We observed obvious improvements across six modalities, with metrics such as FID down to 51.3 from 65.3, IS increasing from 5.7 ± 0.42 to 6.7 ± 0.48 and MS-SSIM improving from 0.14 ± 0.03 to 0.16 ± 0.03 . These findings highlight MINIM's continuous learning potential, driven by the ongoing integration of new medical data.

The objective evaluation for the synthetic brain and breast magnetic resonance images (Fig. 3b, left panel) showed that MINIM can generate high-quality synthetic images in these two domains when mixing these two-modality paired datasets with existing four-modality paired datasets. We presented a comparison of synthetic images generated by MINIM alongside those produced by other generative models, all based on the same textual instructions (Fig. 3c), where we observed that MINIM can generate images with enhanced realism and fidelity.

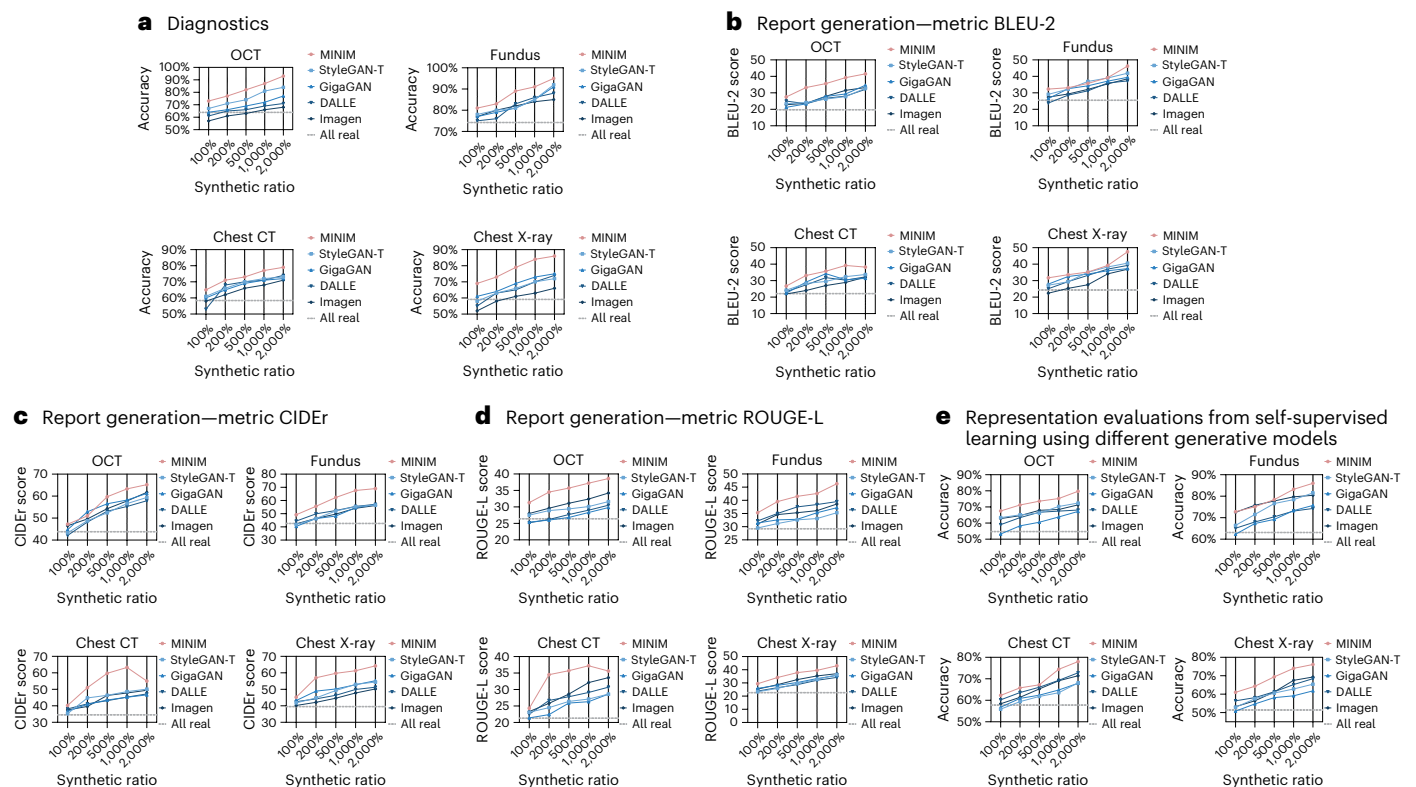


Fig. 4 | Synthetic images serve as data augmentation. a, Performance comparison by incorporating synthetic images from MINIM and other generative models into real data for diagnostic tasks. **b–d**, Performance comparison in three metrics for report generation task by incorporating synthetic images from

MINIM and other generative models into real data. **e**, Performance comparison on the representations from self-supervised learning using MINIM and other generative methods.

More examples conditioned on different descriptions are shown in Extended Data Figs. 5 and 6.

Downstream applications using synthetic images

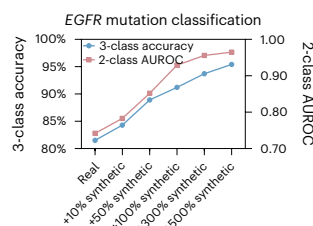
Diagnosis. We demonstrated that adding synthetic data to the training dataset substantially enhances medical diagnosis performance. Multi-class diagnosis models were trained using a Swin Transformer classifier²⁴ with diagnostic labels for each modality. We compared results from models trained on real datasets with those trained on mixed datasets consisting of various proportions of synthetic images. The baseline top-1 classification accuracy using real data exclusively was 0.64 for OCT, 0.74 for fundus, 0.58 for chest CT and 0.59 for chest X-ray. Notably, with a 1:1 ratio of synthetic-to-real data, top-1 classification accuracy improved to 0.73 for OCT, 0.81 for fundus, 0.65 for chest CT and 0.69 for chest X-ray. By increasing the proportion of synthetic data, we observed parallel improvements in top-1 classification accuracy, ultimately reaching average accuracies of 0.93 for OCT, 0.95 for fundus, 0.79 for chest CT and 0.86 for chest X-ray at a 20:1 ratio of synthetic-to-real data (Fig. 4a). To further validate the effectiveness of synthetic data, we specifically synthesized OCT images for classes with poor diagnostic performance. A confusion map visualizing the model's classification results on real data (Extended Data Fig. 4) revealed that the performance for four diagnoses (Macular Hole of Post-Surgery, Choroidal and Retinal Degeneration, Retinal Vasculitis and Endophthalmitis) was substantially worse than others. We subsequently generated 100 tailored OCT images for each of the four diagnoses and invited ophthalmologists to further train the model by selecting typical images. This iterative approach led to marked improvements in diagnostic effectiveness (Extended Data Table 4), illustrating the capacity of synthetic data to augment training datasets and address specific diagnostic challenges. Furthermore, the model successfully diagnosed

atypical cases of common diseases, including retinal vein occlusions (RVOs), choroidal neovascularization (CNV) and diabetic retinopathy (DR), demonstrating its ability to recognize a wide spectrum of disease presentations (Extended Data Fig. 5a).

Report generation. For report generation, we curated text descriptions that appeared more than 10 times in the real dataset, resulting in 60 image reports for each modality. Synthetic images were generated for each report, creating a controlled and more balanced synthetic dataset compared to the real dataset collection process. Using the CLIP+GPT-2 framework²⁵, we trained a report generation model for each modality and evaluated its outputs against clinician-provided annotations using metrics such as BLEU-n, CIDEr and ROUGE-L scores (Fig. 4b–d). Consistent with our diagnostic findings, the combination of synthetic and real data (20:1) improved performance, yielding ROUGE-L scores of 38.6 for OCT, 46.3 for fundus, 35.6 for chest CT and 43.0 for chest X-ray, compared to scores of 26.4, 29.2, 21.3 and 22.6, respectively, for real data alone. Extended Data Fig. 5b presents selected examples of the generated OCT reports, showcasing the model's ability to capture disease information that may be overlooked by ophthalmologists. Moreover, these examples illustrate the potential of synthetic data not only as an augmentation strategy but also as a tool for enriching and enhancing the completeness of medical imaging reports.

Self-supervised learning. Self-supervised learning allows the model to derive meaningful representations from unlabeled data by pre-training on auxiliary targets, a method particularly beneficial in medical imaging, where unlabeled data are abundant and labeling is costly. To assess the effectiveness of our model's synthetic images in self-supervised pre-training, we augmented images from different modalities and employed Siamese network branches²⁶ to learn invariant features.

a Baseline and comparison on *EGFR* mutation classification using different numbers of synthetic images in training datasets



b AI based *EGFR* mutation prediction and its effect on 5-year survival rates of patients with advanced lung cancer

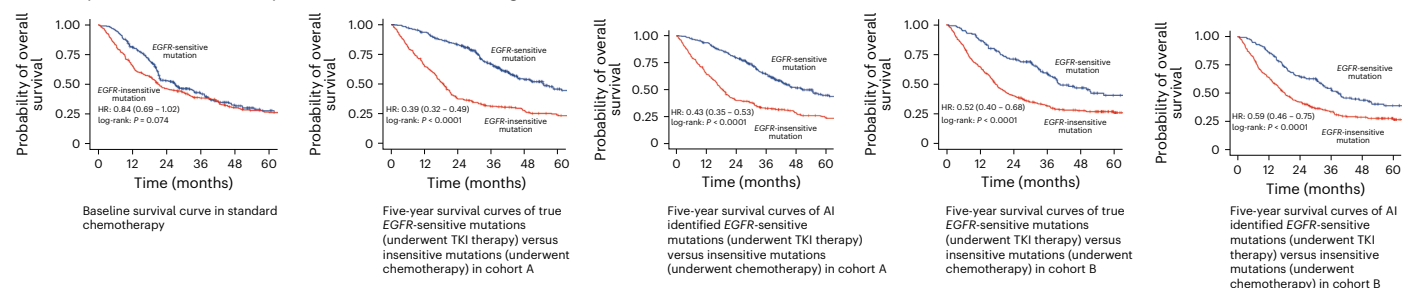


Fig. 5 | AI-based mutation prediction performance and its effect on 5-year survival rates of patients with advanced lung cancer. a, Baseline and performance comparison on *EGFR* mutation predictions using different numbers of synthetic images in training datasets. **b**, Left to right: a baseline survival curve in patients who underwent standard chemotherapy; 5-year survival curves of true *EGFR*-sensitive mutations (underwent TKI therapy) versus *EGFR*-insensitive mutations (underwent chemotherapy) in cohort A; 5-year survival curves of AI identified *EGFR*-sensitive mutations (underwent TKI therapy) versus

c *HER2* mutation detection using real data and mixed data

Train:real test:real					Train:mixed test:real				
Actual \ Predicted	No tumor	Tumor with <i>HER2</i>	Tumor without <i>HER2</i>	Sum	Actual \ Predicted	No tumor	Tumor with <i>HER2</i>	Tumor without <i>HER2</i>	Sum
No tumor	333	24	39	396	No tumor	387	6	9	402
				84.09%					96.27%
Tumor with <i>HER2</i>	23	127	18	168	Tumor with <i>HER2</i>	5	157	6	168
				75.60%					93.45%
Tumor without <i>HER2</i>	37	24	169	230	Tumor without <i>HER2</i>	13	9	208	230
				73.48%					80.43%
Sum	393	175	226	629/794	Sum	405	172	223	752/800
				79.22%					94.00%

	Accuracy			F1-score			Precision			Recall		
	All	No tumor	Tumor with <i>HER2</i>	Tumor without <i>HER2</i>	All	No tumor	Tumor with <i>HER2</i>	Tumor without <i>HER2</i>	All	No tumor	Tumor with <i>HER2</i>	Tumor without <i>HER2</i>
Real	0.788	0.847	0.736	0.738	0.833	0.756	0.734	0.841	0.717	0.717	0.741	0.741
Real+synthetic	0.940	0.959	0.923	0.918	0.962	0.934	0.904	0.955	0.912	0.912	0.932	0.932

EGFR-insensitive mutations (underwent chemotherapy) in cohort A; 5-year survival curves of true *EGFR*-sensitive mutations (underwent TKI therapy) versus *EGFR*-insensitive mutations (underwent chemotherapy) in cohort B; 5-year survival curves of AI identified the *EGFR*-sensitive mutations (underwent TKI therapy) versus *EGFR*-insensitive mutations (underwent chemotherapy) in cohort B. **c**, *HER2* mutation prediction performance by adding synthetic images into the training dataset.

The DenseNet-121 model²⁷ was pre-trained on auxiliary tasks, which involved fine-tuning only the classification layer from 100% to 2,000% of the available training data. Our findings indicate that models fine-tuned with both synthetic and real images consistently outperformed those fine-tuned exclusively on real images, with top-1 accuracy improving substantially, achieving 79.7 for OCT, 86.0 for fundus, 78.0 for chest CT and 76.1 for chest X-ray, compared to the initial accuracies of 54.7, 63.1, 57.8 and 51.4, respectively (Fig. 4e).

Clinical applications using synthetic images

***EGFR* mutation detection in lung cancer.** We investigated whether synthetic images can improve the top-1 accuracy of *EGFR* (<https://www.ncbi.nlm.nih.gov/gene/1956>) mutation classifications using chest CT scans (Fig. 5a). A Swin Transformer was employed to classify real CT images into three *EGFR* mutation types: wild, sensitive and resistant. Baseline performance using only real data achieved an average top-1 accuracy of 81.5%. By progressively incorporating synthetic data into the training set, we observed substantial improvements in classification accuracy. Remarkably, a 1:1 ratio of synthetic-to-real data for training increased classification accuracy to 91.2%, with further gains reaching 95.4% when the synthetic-to-real data ratio reached 5:1. Additional increases in synthetic data beyond this point did not yield noticeable improvements. In a subsequent binary classification task, combining *EGFR* wild-type and resistant-type mutations, the inclusion of synthetic data similarly enhanced model performance using the area under the receiver operating characteristic curve (AUROC). AUROC improved from 74.2% with real data alone to 96.5% when synthetic data were expanded five-fold. These trends underscore the value of synthetic data in improving the predictive accuracy of AI models for *EGFR* mutation detection in lung cancer.

We further validated these results in a retrospective clinical study using two independent cohorts of patients (Methods). We evaluated

our model's prediction of patients with advanced lung cancer, based on *EGFR* mutations, who could benefit from tyrosine kinase inhibitor (TKI)-targeted therapy. We first determined the baseline survival curve of patients who underwent standard chemotherapy (Fig. 5b). Then, using our model's predictions, we identified patients with *EGFR*-sensitive mutations (who received TKI therapy) and compared their survival rates to those with *EGFR*-resistant mutations (who received chemotherapy). In the first group of patients (cohort A), *EGFR*-sensitive mutations had a 29.5% higher 5-year overall survival (OS) (*EGFR*-sensitive mutations, 53.4% (95% confidence interval (CI): 47.5–60.1%); *EGFR*-resistant mutations, 23.9% (95% CI: 19.3–29.5%)) and a longer median survival time (*EGFR*-sensitive mutations, 54.7 months (95% CI: 45.4–NA); *EGFR*-resistant mutations, 18.8 months (95% CI: 16.6–21.1)) (Fig. 5b). The 5-year progression-free survival (PFS) and objective response rate (ORR) also improved from 12.2% to 41.6% (Extended Data Fig. 2a, b) and 12.3% to 30.9% (Extended Data Fig. 3a, b), respectively. We confirmed similar results in the second patient group (cohort B), with a 14.7% increase in 5-year OS (*EGFR*-sensitive mutations, 41.0% (95% CI: 32.2–52.2%); *EGFR*-resistant mutations, 26.3% (95% CI: 21.6–32.1%)) and longer median survival time (*EGFR*-sensitive mutations, 40.6 months (95% CI: 36.4–63.2); *EGFR*-resistant mutations, 18.1 months (95% CI: 16.2–21.0)) (Fig. 5b). Error rates of our model's predictions for both cohort A (*EGFR*-sensitive mutations, 0.12 (4/327), *EGFR*-resistant mutations, 0.018 (6/321)) and cohort B (*EGFR*-sensitive mutations, 0.028 (4/142), *EGFR*-resistant mutations, 0.04 (6/316)) were low. The PFS and ORR also improved from 19.1% to 44.7% (Extended Data Fig. 2c, d) and 18.9% to 33.1% (Extended Data Fig. 3c, d), respectively.

***HER2* status detection in breast cancer.** Given the complexity of *HER2* status and its critical role in determining treatment outcomes in *HER2*-targeted therapies, our goal was to explore whether integrating

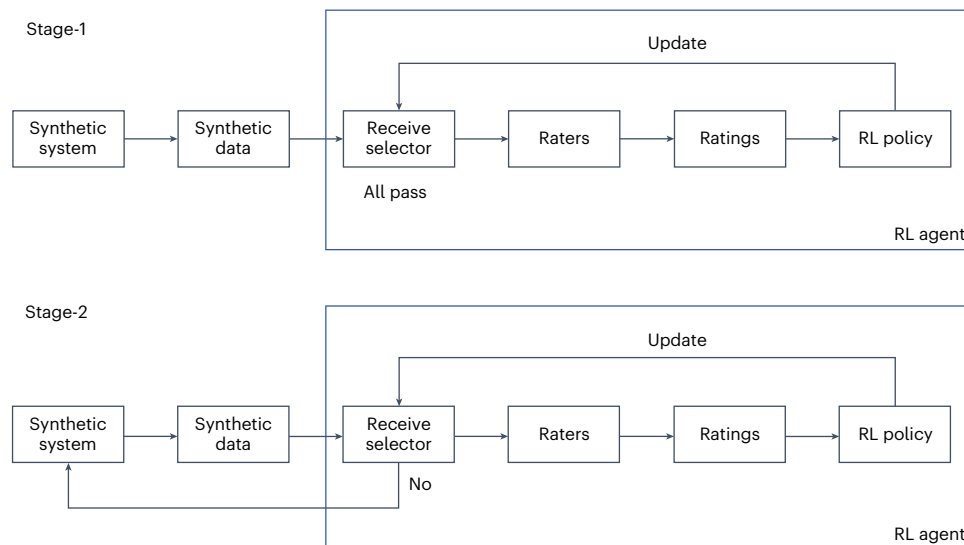


Fig. 6 | A two-stage RL strategy using human ratings. The first stage involves the generation of synthetic images from a generative model. The generated images are then passed to a 'Receive Selector', which operates as a classification module.

The second stage introduces an active selection process. Again, synthetic images are fed into the system, but, this time, the 'Receive Selector' classification module decides which images to pass based on the classification results.

synthetic images from our model could enhance the accuracy of HER2 status classification. We first established a baseline that was trained exclusively on real datasets, achieving a classification accuracy of approximately 79.2% for distinguishing among three classes (no tumor/tumor with HER2/tumor without HER2) (Fig. 5c, left panel). With the progressive addition of synthetic images with text descriptions at ratios of 2, 3, 5 and 10 times the proportion of real images, we observed substantial improvements in the classification accuracy, peaking at 94.0% when using a synthetic-to-real image ratio of 10:1 (Fig. 5c, right panel).

Discussion

Here we present a unified medical image–text generative model (MINIM) that was trained on paired medical images with textual descriptions across various modalities and organs (OCT, fundus, chest X-ray, chest CT, brain MRI and breast MRI). We demonstrated MINIM's capability to generate high-quality synthetic images for each organ and imaging modality given textual descriptions. The generated data can function as a supplementary resource for the development of AI models, particularly in scenarios where there is a lack of sufficient data due to privacy or cost^{28,29}. Our findings indicate that AI models trained on both real and synthetic data exhibit improved predictive capabilities, with this enhancement being very beneficial in the context of rare disease diagnoses, report generation and self-supervised learning. On clinical grounds, we demonstrated that MINIM can substantially enhance the detection of HER2 mutations in the breast. Furthermore, we applied MINIM to an *EGFR* mutation target therapy in a clinical study and demonstrated improved 5-year survival rates in two independent cohorts.

Our evaluations of classification accuracy in lung cancer and breast cancer aim to demonstrate the clinical relevance and advantages of our AI model. For lung cancer, we focused on *EGFR* mutation classification because of its critical role in determining the suitability of TKI-targeted treatments, which substantially improves survival in patients with advanced non-small cell lung cancer^{30,31}. By evaluating the model's performance in both three-class (wild, sensitive and resistant) and binary (sensitive versus resistant+wild) classification tasks, we demonstrated the model's robustness in accurately predicting *EGFR* status, which is crucial for guiding targeted therapy decisions. Our AI-driven approach provides a non-invasive, efficient and scalable alternative to traditional biopsy-based *EGFR* mutation detection, ultimately improving patient outcomes and expanding access to precision medicine.

Breast cancer is the most common cancer in women worldwide, posing a considerable threat to women's health³². HER2 is a critical biomarker for determining the benefit of HER2-targeted therapies³³. Traditionally, only HER2 amplification or overexpression was used to predict an enhanced survival benefit from HER2-targeted therapy³⁴, and our model addresses this limitation by offering a more comprehensive and nuanced approach to HER2 status classification. By leveraging both real and synthetic datasets, our model identifies subtle patterns associated with HER2-positive tumors that may not be detectable through conventional methods. This enhanced ability to classify HER2 status more accurately could lead to better identification of patients who would benefit from HER2-targeted therapies, ultimately improving treatment outcomes and survival rates in patients with HER2-positive breast cancer.

Despite the advancements in GANs, challenges persist in generating high-quality medical images due to limited training data³⁵. MINIM addresses this limitation by integrating both textual and imaging data, effectively expanding the input space and enhancing the quality of the generated images. Furthermore, concerns regarding the clinical relevancy of generated images³⁶ are also mitigated by training the model with paired image and text inputs, ensuring that the outputs are closely aligned with the clinical contexts described in the accompanying text.

Another longstanding question in the field of generative AI has been the extent to which synthetic data could enhance predictive model performance^{5,37,38}. Here we provide evidence that the proposed generative model can self-improve through RLHF and transfer learning (Fig. 1a), enabling continuous learning and addressing the challenge of adapting to new data domains³⁹. These findings have major implications for the development of generative AI models that can adapt and learn prospectively over time across various applications^{40–42}.

We recognize several limitations in our study that warrant attention. First and foremost, the datasets employed are of a relatively small scale and predominantly derived from Caucasian and Chinese populations. This limited demographic scope, coupled with the study's moderate size sourced from published datasets, could potentially lead to biases. To mitigate this, future iterations of the model should incorporate data from a more varied array of sources, ensuring a comprehensive representation of different genders and ethnic groups. This expansion would enhance not only the model's robustness but also its applicability across a broader demographic spectrum.

Another important limitation lies in the constraints imposed by the token length of the CLIP text encoder⁴³. Currently, our training

is confined to sections of OCT and chest X-ray reports, which may inadvertently overlook critical image feature information. To address this, we propose the integration of more detailed textual information, including findings and clinical details. This enrichment will enhance the model's conditioning capabilities, allowing for a more nuanced and comprehensive analysis of medical images in conjunction with their accompanying textual reports.

A third area of concern is the challenge of aligning text and images, especially when dealing with longer prompts⁴⁴. This alignment difficulty could potentially be improved through innovative approaches, such as employing reward models that are trained based on human feedback. Such methods would likely increase the precision of text–image alignment, making the model more effective in handling complex, lengthy prompts. In addition to these challenges, there is also the risk of the model overfitting. To counteract this, we emphasize the need for meticulous strategies, such as integrating adversarial loss and adaptively managing the learning rate during fine-tuning, especially under the constraints of limited data⁴⁵. Continuous human oversight and feedback are crucial components of this process, ensuring that the model remains accurate and reliable.

Online content

Any methods, additional references, Nature Portfolio reporting summaries, source data, extended data, supplementary information, acknowledgements, peer review information; details of author contributions and competing interests; and statements of data and code availability are available at <https://doi.org/10.1038/s41591-024-03359-y>.

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Methods

Dataset descriptions and ethical approvals

We collected OCT data and reports from multiple medical institutes to build an extensive dataset (Extended Data Table 1, left). We combined image–report pairs from China Eye Image Investigation and randomly split the data based on patient identifications to build the training and validation sets (11,438 data pairs). We used data from the Eye Hospital of Wenzhou Medical University for external validation (2,287 data pairs). In addition to OCT images with reports from public datasets⁴⁶, we assembled a high-quality dataset comprising 13,725 paired image–text OCT samples.

The retinal fundus images were collected from the study cohorts of the China Bioage Investigation Consortium (Extended Data Table 1, middle), including the northern China cohort, which is from the China Suboptimal Health Cohort Study in Tangshan City, Hebei Province, China, and the southern China cohort, which is from the Nanfang Hospital in Guangzhou, Guangdong Province, Zhuhai People's Hospital. A total of 59,420 image–text pairs consented to participate in our study.

The chest CT and X-ray images were collected from West China Hospital and Guangzhou Medical University (Extended Data Table 1, right). For each patient with his/her medical record, we selected 10 CT images and 10 X-ray images, resulting in 74,900 paired data for 3,745 patients.

The brain MRI dataset was collected from multiple centers, including 1,960 patients from 16 hospitals in China⁴⁷. These patients collectively contributed a total of 30,800 tumor-annotated images, including 13,859 T1-channel images and 16,941 T2-channel images.

The breast tumor imaging dataset was obtained from the Guangzhou National Laboratory and Sun Yat-sen University Memorial Hospital (as shown in Extended Data Fig. 6), encompassing 3,694 patients diagnosed with breast MR images. These patients provided a total of 34,506 annotated tumor images and 35,000 normal breast images, which included 16,017 T1-channel images, 6,129 T2-channel images and 47,360 contrast-enhanced T1-channel images. Notably, 3,200 images contain information on the HER2 gene, which is specifically used to train the model to associate HER2 gene expression with MRI characteristics. The remaining images serve as an additional dataset to train the stable diffusion model, ensuring robust and reliable performance.

Approvals for the research were secured from the institutional review boards/ethics committees of all participating hospitals or institutes. Consent forms were signed by all involved patients. The study was carried out in accordance with the principles of the Declaration of Helsinki and conformed to the policies of the Chinese Center for Disease Control and Prevention as well as the Chinese Health Laws.

Image grading from clinicians

OCT and fundus images went through a tiered grading system consisting of multiple layers of trained graders of increasing expertise for verification and correction of image labels. Each image imported into the database started with a label matching the most recent diagnosis of the patient and examined each paired report. The first tier of graders consisted of medical students and ophthalmology residents who had taken and passed an interpretation course review. This first tier of graders conducted initial quality control on real patient images and excluded images containing severe artifacts or noticeable image resolution reductions. The second tier of graders consisted of five ophthalmologists who had at least 10 years of retinal subspecialty practice, and they independently graded each image and corresponding report. The presence or absence of images reflecting features of vitreous, retinal, retinal pigment epithelial (RPE) and choroid and various retinal disease diagnoses, including choroidal neovascularization (active or in the form of subretinal fibrosis), macular edema, drusen, RVOs, epiretinal membrane and other pathologies visible on the OCT scan, were recorded. Finally, a third tier of two senior independent retinal specialists, each with over 20 years of clinical retina experience, verified the true labels and corresponding reports for each image. To

account for human error in grading, a validation subset of 993 scans was graded separately by two ophthalmologist graders, with disagreement in clinical labels arbitrated by a senior retinal specialist. For chest X-ray, chest CT, brain MRI and breast MR images, all modalities were initially screened for quality control by removing all low-quality or unreadable scans. Similar to previous ophthalmological image grading, the diagnoses here were then graded by three radiology expert physicians who have at least 10 years of practice experience. To account for any grading errors, the evaluation set was also checked by another expert.

In assessing AI-based *EGFR* mutation status predictions, we employed two independent retrospective cohorts of patients with advanced lung cancer (stage III or stage IV) from West China Hospital in Chengdu, China, or Guangzhou Medical University First Affiliated Hospital, respectively. A part of the descriptions of these two cohorts was published^{48,49}. We used patients from a period up to 2014 for training. We used consecutive patients from 2015–2020 for a 5-year survival evaluation (cohort A: West China Hospital; cohort B, Guangzhou Medical University First Affiliated Hospital). We used Kaplan–Meier survival curves to compare survival rates between patients with *EGFR*-sensitive mutation versus patients with *EGFR*-insensitive mutation. All patients received standard chemotherapy plus TKI therapy if they had undergone a tumor tissue biopsy and *EGFR* gene sequencing to determine the *EGFR* mutation status. An evaluation committee consisting of three senior oncologists gave blinded and de-identified cohort A and cohort B patients to a computer scientist for AI-based *EGFR* mutation status prediction.

MINIM's framework and implementation details

MINIM is a novel framework designed for the generation of medical images through a diffusion model, leveraging paired medical images $I = \{i_1, i_2, i_3, \dots, i_n\}$ and their corresponding textual descriptions $D = \{d_1, d_2, d_3, \dots, d_n\}$ across different medical image modalities $M = \{m_1, m_2, m_3, \dots, m_k\}$. To facilitate the accurate generation of specific modality images, we concatenated the modality information m_k with the description information d_n to generate the text input $[m_k; d_n]$ of MINIM. During the training phase, MINIM employs a BERT tokenizer, E_{BERT} , to encode both the modality information and description information separately. The framework introduces a series of random Gaussian noises to the input image progressively, following the equation:

$$q(i_t|i_{t-1}) = N\left(i_t; \sqrt{1 - \beta_t}i_{t-1}, \beta_t I\right),$$

where β_t is a linear noise scheduler defined as $\beta_t = \beta_{\min} + t \times \frac{\beta_{\max} - \beta_{\min}}{T}$, and i_T is pure noise input. Then, MINIM learns to reverse the diffusion process using a U-Net architecture with cross-attention mechanisms on both modality information and description information:

$$p_{\theta}(i_{t-1}|i_t, E_T) = N\left(i_{t-1}; \mu_{\theta}(i_t, t, E_T), \sum_{\theta}(i_t, t)\right),$$

where the μ_{θ} is predicted by the output of the cross-attention between input texts and images, which incorporates cross-attention mechanisms at different U-Net layers. In the shallow layers of the U-Net encoder and decoder, MINIM applies cross-attention between image embedding $U_{\text{shallow}}(i_t)$ and modality information embeddings $E_{\text{BERT}}(m)$, and, in the deep layers of the U-Net encoder and decoder, MINIM applies cross-attention between image embedding $U_{\text{deep}}(i_t)$ and modality information embeddings $E_{\text{BERT}}(d)$. The cross-attention function is defined as:

$$\text{Cross Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V,$$

where Q is derived from the U-Net features, and K and V are derived from the corresponding text embeddings ($E_{\text{BERT}}(m)$ or $E_{\text{BERT}}(d)$).

During inference, the generation of images occurs through an iterative denoising process starting from random noise i_T .

$$i_{t-1} = \mu_\theta(i_t, t, E_T) + \sigma_t z,$$

where $z \sim \mathcal{N}(0, \mathbf{I})$ and $\sigma_t^2 = \beta_t$. This sequential cross-attention approach allows MINIM to first incorporate modality-specific information and then refine the features with more detailed descriptive information. By concatenating modality and description information, the model can learn to distinguish between general modality characteristics and specific image details described in the text, potentially leading to more accurate and controllable medical image generation.

To optimize MINIM for generating medical images, we used pre-trained U-Net parameters sourced from the general domain and fine-tuned the model with four distinct modalities of medical images, each paired with its corresponding textual description. All codes were implemented in PyTorch. The fine-tuning experiments were conducted on a high-performance computing cluster equipped with eight NVIDIA A100 GPUs. We optimized the proficiency in generating medical images by adjusting hyperparameters, initializations and epochs. All the input images are rescaled at a resolution of 512×512 pixels and processed in batches of 128. In this configuration, fine-tuning a model required approximately 10 h for 1,000 training steps and 4 d for 10,000 steps. We obtained the U-Net model weights for the SD pipeline (version 1.4) and CLIP (ViT-large) from HuggingFace. Our implementations used the ‘transformers’ and ‘diffusers’ libraries. We disabled the built-in safety checker during the generation process owing to its high false-positive rate in response to medical prompts. For synthesizing medical images, we employed a classifier-free guidance scale set at 4.0 and set the inference steps at 100, using the default noise scheduler, which involves the pseudo-numerical methods for diffusion models. The best-performing model was selected as our generative model of choice (Fig. 1a).

Comparative methods and ablation study

We compared MINIM with the latest text-to-image generative models in the AI community, including Imagen¹⁸, DALL-E¹⁹, GigaGAN¹⁷ and StyleGAN-T¹⁶. Pre-trained models on their released checkpoints were fine-tuned on our dataset. Hyperparameters were employed following the instructions in their original papers. We performed an ablation study to select the optimal hyperparameters for MINIM. The quality of the synthetic images was assessed using the average of three metrics (FID, IS and MS-SSIM) on all datasets (Extended Data Table 2). The results demonstrate that, as the number of training steps increased, the quality of synthetic images improved substantially, with FID decreasing to 57.91, MS-SSIM improving to 0.18 and IS increasing to 5.89 at 20,000 training steps, which was the choice in our further experiments. We also examined the impact of different model initializations and image resolutions. Randomized weight initialization of U-Net and CLIP models led to poorer performance, as evidenced by higher FID scores. In addition, generating lower-resolution images (256×256) resulted in worse generative quality. In summary, the optimal performance was achieved by the full MINIM model, initialized with pre-trained weights and trained on 512×512 image resolution for 20,000 steps. This configuration showed the best balance of the three metrics. Extended Data Table 3 presents a comparison of the classification performance using different configurations in diagnosing various eye conditions using synthetic OCT images. The table highlights the improvement in accuracy as the number of training steps increases. Initially, the original SD model had a random accuracy of 0.51. After 10,000 training steps, the model achieved substantial improvements in accuracy: 0.82 for CNV classification, 0.84 for DME, 0.88 for DRUSEN and 0.91 for Normal cases. Furthermore, the table investigates the impact of different model initializations and image resolutions on the classification performance. Randomized weight initialization of U-Net and CLIP models, as well as lower-resolution images (256×256), led to

suboptimal results compared to the full MINIM model. The full MINIM model, trained for 20,000 steps, exhibited an exceptional average accuracy of 0.94, closely matching its performance on real data. This demonstrates the robustness and reliability of the MINIM model in generating high-quality synthetic images suitable for accurate multi-class diagnosis.

RL implementation details

We introduce a two-stage RLHF framework that employs synthetic data to iteratively train an agent (Fig. 6). The framework is designed to harness the advantages of synthetic images and radiologist ratings, allowing for the generation of high-quality data, which is especially beneficial in situations where real-world data are scarce, expensive or risky to obtain. The first stage involves the generation of synthetic images from the MINIM model. The generated images are then passed to a receive selector, which operates as a classification module. In this stage, the classification module is designed to let all instances pass through without selection. Subsequently, human raters evaluate the synthetic data, and their assessments are encoded as ratings from 1 to 3, where 1 stands for ‘The image quality is too low’; 2 stands for ‘The image quality meets the standard, but it does not correspond with the report’; and 3 stands for ‘The image quality meets the standard and is generally consistent with the report’. These ratings serve to update the RL policy, thereby closing the loop of this stage. The RL policy π is updated according to the following equation:

$$\pi^{new} = \pi + \alpha \sum_{s,a} (R(s,a) - \pi(s,a)) \nabla \pi(s,a)$$

where $R(s,a)$ denotes the rating for the state–action pair (s,a) , and α is the learning rate.

The second stage introduces an active selection process. Again, synthetic images are fed into the system, but, this time, the ‘Receive Selector’ classification module decides which images to pass based on the classification results. Only the selected images are forwarded to the human raters for evaluation. The non-selected instances are discarded. The ratings from the human raters update the RL policy, similar to the first stage but with a different learning rate. The selective filtering focuses the policy update on higher-quality images, which is crucial for improving learning efficiency and effectiveness. The updated policy is represented by:

$$\pi^{new} = \pi + \beta \sum_{s,a \in S_{selected}} \frac{P_{selected}(x_i)}{P_\pi(a|s)} (R(s,a) - V_\pi(s)) \nabla \pi(s,a)$$

where β is a different learning rate from α , reflecting the different data distributions between the two stages. $P_{selected}(x_i)$ is the probability that the instance x_i is selected by the ‘Receive Selector’; $P_\pi(a|s)$ is the probability of taking action in state s under the current policy π ; and $V_\pi(s)$ is the value function under policy π for state s . In this formulation, the term $R(s,a) - V_\pi(s)$ represents the temporal difference error, a measure of how the outcome was compared to the agent’s expectations⁵⁰.

Implementations for downstream analysis

To evaluate the effectiveness of synthetic data in diagnosis, we selected the category labels that appeared more than five times in the training set. We then used the Swin Transformer model to train a multi-class classification diagnostic model on the corresponding images and labels. First, the synthetic images of an equal number to the training images were generated using fine-tuned MINIM for each text description in the training dataset, and they were mapped to the corresponding diagnosis of the text. These images were then incorporated as additional inputs into the Swin Transformer classification model during its training phase. For the evaluation of the generated diagnosis results, we used Accuracy, F1 score and AUC as evaluation metrics. In a further

scaling-up experiment, the number of synthetic images generated varied from 100%, 200%, 500%, 1,000% and 2,000%, with the results of each variation reported accordingly.

The experimental settings on report generation using synthetic data for each image modality were as follows. Text descriptions with a frequency exceeding 10 were chosen, along with their corresponding images, forming real datasets. These text descriptions were subsequently used for generating synthetic images, which served as additional inputs for the CLIP+GPT2-based image captioning model during its training phase. Subsequently, an independent test dataset was employed to produce corresponding reports, which were compared against the ground truth. For the evaluation of the generated reports compared to the ground truth reports, we employed established natural language generation metrics. These metrics included BLEU with 1–4 g, CIDEr and ROUGE-L. In a further scaling-up experiment, the number of synthetic images generated varied from 100%, 200%, 500%, 1,000% and 2,000%, with the results of each variation reported accordingly.

In our assessment of self-supervised learning methodologies using synthetic data, we employed a Siamese network architecture. For each image modality, we introduced two distinct augmentations of a synthetic image into separate branches of the network. The self-supervised pre-trained models of DenseNet-121 underwent linear probing, with only a classification layer fine-tuned on 100%, 200%, 500%, 1,000% and 2,000% of the training data while keeping other model weights fixed. Top-1 classification accuracy was reported using a mixture of real data and different numbers of synthetic data and real data only (Fig. 4e).

Implementation details of *EGFR* mutation type detection

We conducted two-class and three-class *EGFR* mutation type classifications, comparing a baseline model that used only real images with several models that combined real images with varying proportions of synthetic data, ranging from 10% to 500%. We used Swin Transformer to achieve an end-to-end architecture that encourages the model to predict a class probability. Training of the models by back-propagation of errors was performed in batches of 32 images resized to pixels for 50 epochs with a learning rate of 1×10^{-4} . The training was performed using the Adam optimizer with a weight decay of 1×10^{-5} . Transformations of random horizontal and vertical flips were added to each batch during training as data augmentation to enable improved and generalized network learning. The models selected for evaluation on test sets were the models with the best validation loss on the training set.

Implementation details of *HER2* status detection

We trained a three-layer convolutional neural network (CNN) to classify breast MR images into three classes: no tumor, tumor with *HER2* positive and negative status. The model comprises three convolutional layers, each followed by a rectified linear unit (ReLU) activation function and a maximum pooling layer. At the end of the model, there are three linear functions separated by ReLU activations. We first trained a basic classifier on a dataset composed of solely real data (3,200 images). Among these, there were 1,598 images without tumors, 687 images containing tumors with *HER2*-positive status and 915 images containing tumors with *HER2*-negative status. Using the same architecture, we then trained a new classifier on a hybrid dataset consisting of both real and synthetic data. This dataset included all of the aforementioned real data as well as 10 times of synthetic images. Within synthetic images, there were 16,274 images without tumors, 7,582 images containing tumors with *HER2*-positive status and 6,292 images containing tumors with *HER2*-negative status. During training, we employed the Adam optimizer with a loss function of CrossEntropy. We set the learning rate to a fixed value of 0.001 and trained for a total of 20 epochs using a batch size of 128. The test results of these two classifiers are depicted in Fig. 5c.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

Restrictions apply to the availability of the developmental and validation datasets, which were used with permission of the participants for the present study. De-identified data may be available for research purposes from the corresponding authors upon reasonable request.

Code availability

The code to reproduce the results can be accessed at <https://github.com/WithStomach/MINIM>.

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Author contributions

J.W., K.W., Y.Y., Y.L., W.X., Z.S., F.L., Z.Z., Y.G., L.Y., H.-Y.Z., H.M., W.Z., L.H., L.Z., R.G., I.C., B.D., L.C., X.C., J.L., M.-H.Z., D.B.-H., O.M., M.L., Y.K., J.L., S.Z., T.G., J.Z., K.X., E.O., H.L., Y.Y., K.Z. and J.Q. collected and analyzed the data. K.Z. and J.W. conceived and designed the project. K.Z., J.W. and J.Q. designed experiments and supervised the project. J.W. and K.Z. wrote the manuscript. All authors discussed the results and reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Extended data is available for this paper at <https://doi.org/10.1038/s41591-024-03359-y>.

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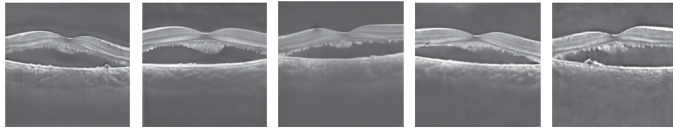
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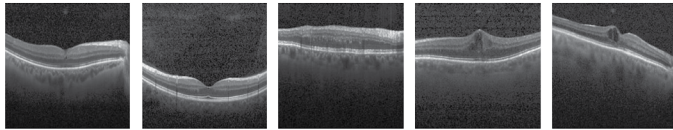
peer review of this work. Primary Handling Editor: Joao Monteiro, in collaboration with the *Nature Medicine* team.

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prompt: "OCT: Part of the pigment epithelium can be seen to be bulging, and there is no obvious abnormality in the shape of the macula."



prompt: "OCT: The reflective strips on the retinal surface are not smooth, the neuroepithelial layer in the macular zone is thickened, punctate medium-high reflections can be seen between the retinal layers, and cystic low-reflection areas can be seen between the retinal layers."



prompt: "X-ray: Lobar pneumonia."



prompt: "CT: Squamous cell carcinoma of the lungs, postoperative, EGFR sensitive type."



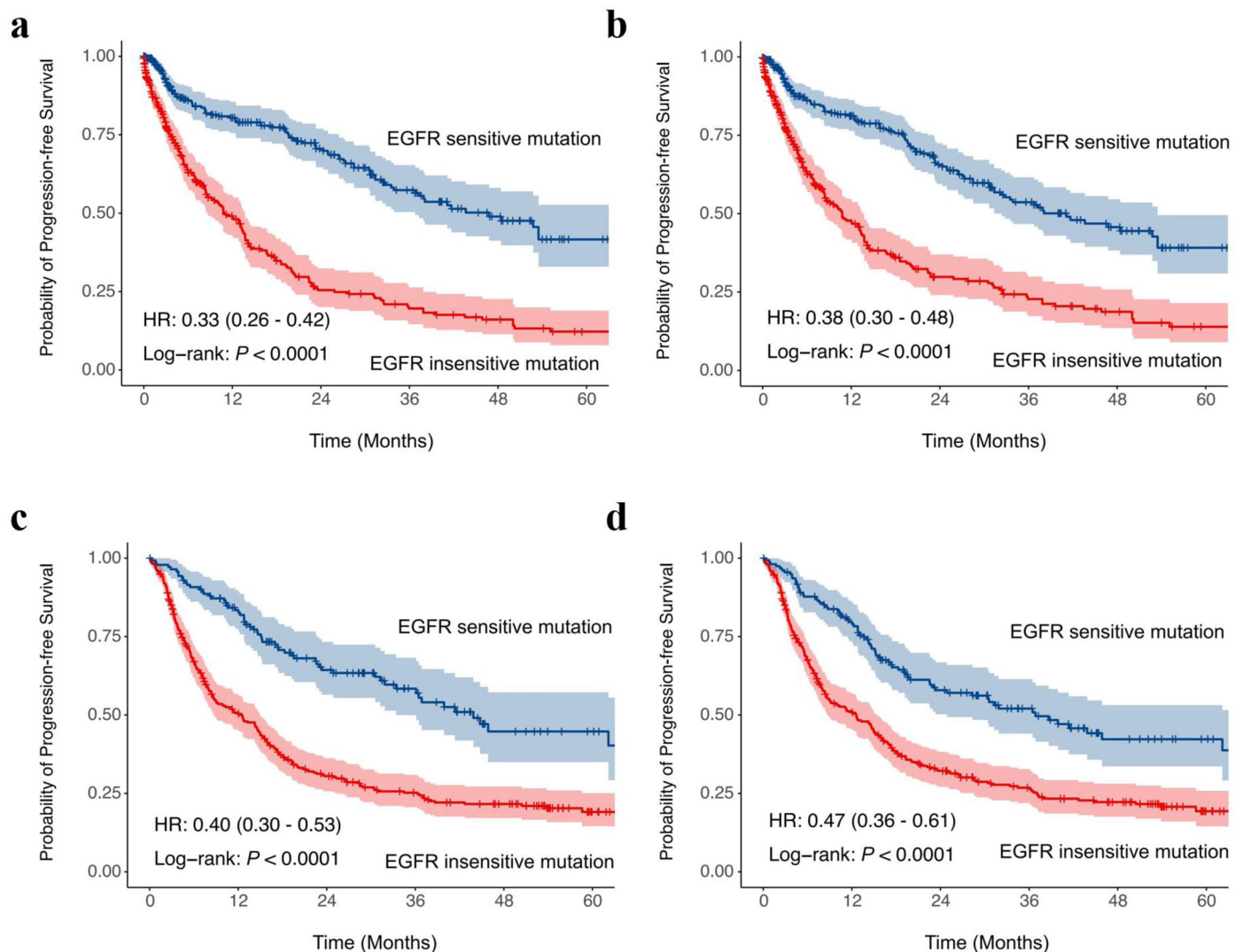
prompt: "MRI: breast tumor with HER2 mutation from the view of T1c."



prompt: "MRI: brain gliomas."

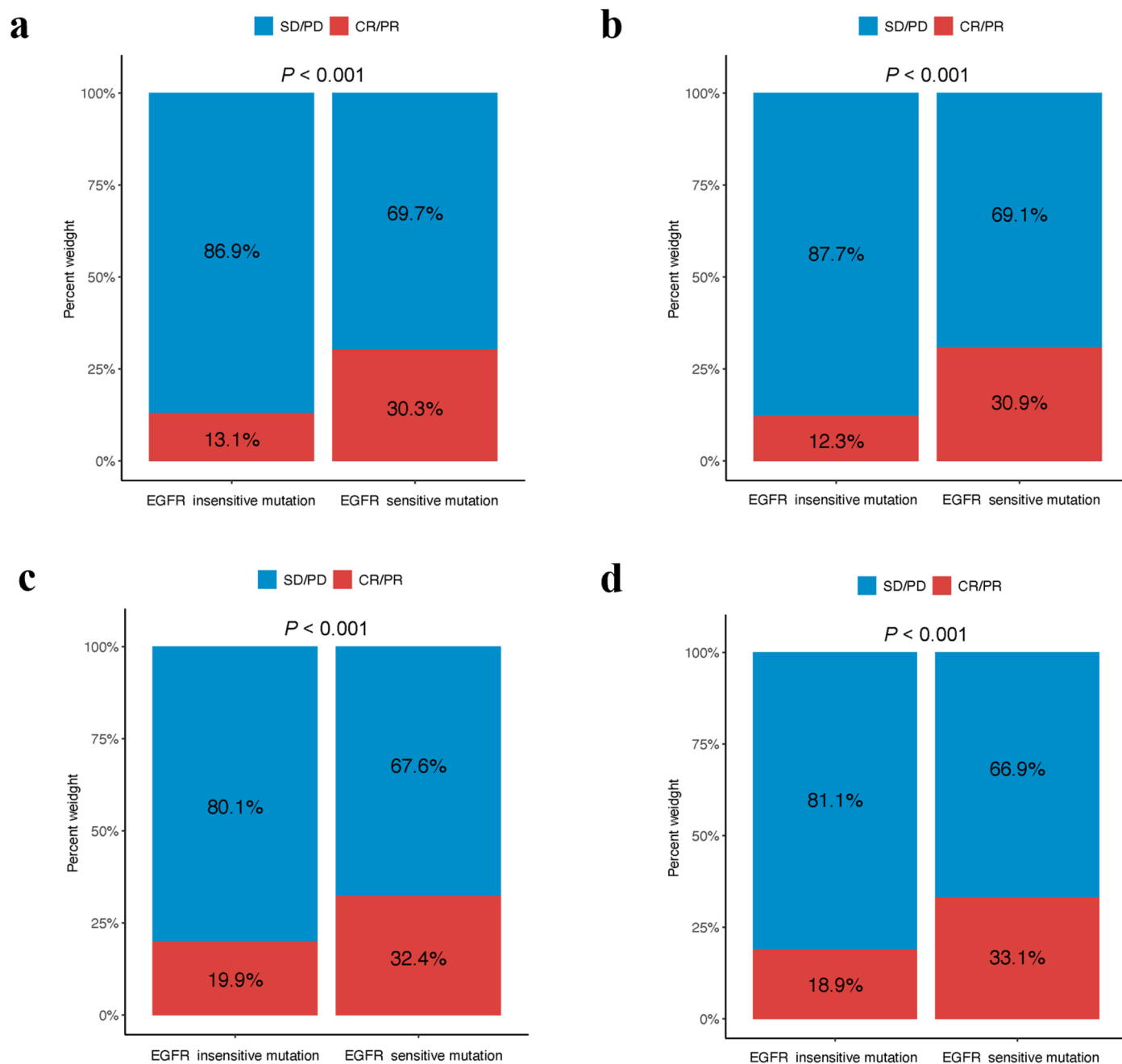


Extended Data Fig. 1 | Text-conditioned synthesis of medical images. Samples were created by prompting a fine-tuned model using annotations made by ophthalmologists and radiologists.



Extended Data Fig. 2 | AI-based *EGFR* mutation prediction and its effect on progression-free survival (PFS) rates of advanced lung cancer patients. **a**, PFS curves of true *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in the cohort A. **b**, PFS curves of AI identified *EGFR* sensitive mutations (underwent TKI therapy) versus

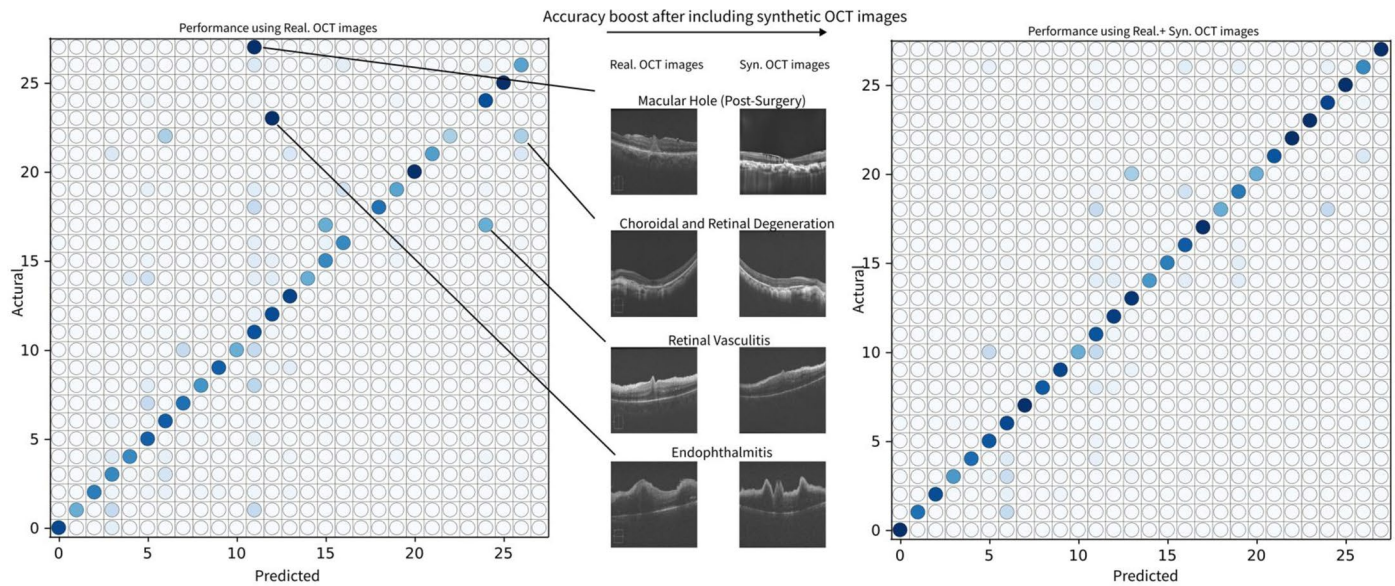
insensitive mutations (underwent chemotherapy) in cohort A. **c**, PFS curves of true *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in cohort B. **d**, PFS curves of AI identified *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in cohort B.



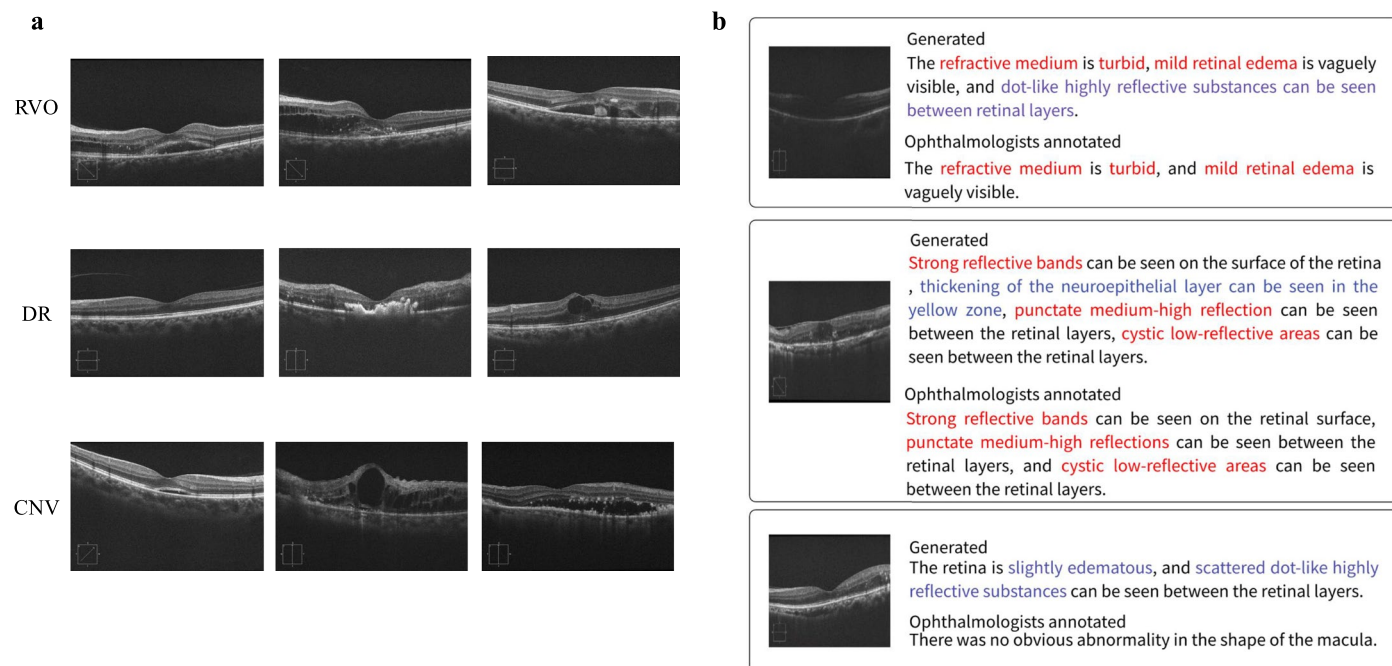
Extended Data Fig. 3 | AI-based *EGFR* mutation prediction and its effect on objective response rate (ORR) of advanced lung cancer patients.

a, ORR of true *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in the cohort A. **b**, ORR of AI identified *EGFR* sensitive mutations (underwent TKI therapy) versus

insensitive mutations (underwent chemotherapy) in cohort A. **c**, ORR of true *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in cohort B. **d**, ORR of AI identified *EGFR* sensitive mutations (underwent TKI therapy) versus insensitive mutations (underwent chemotherapy) in cohort B.

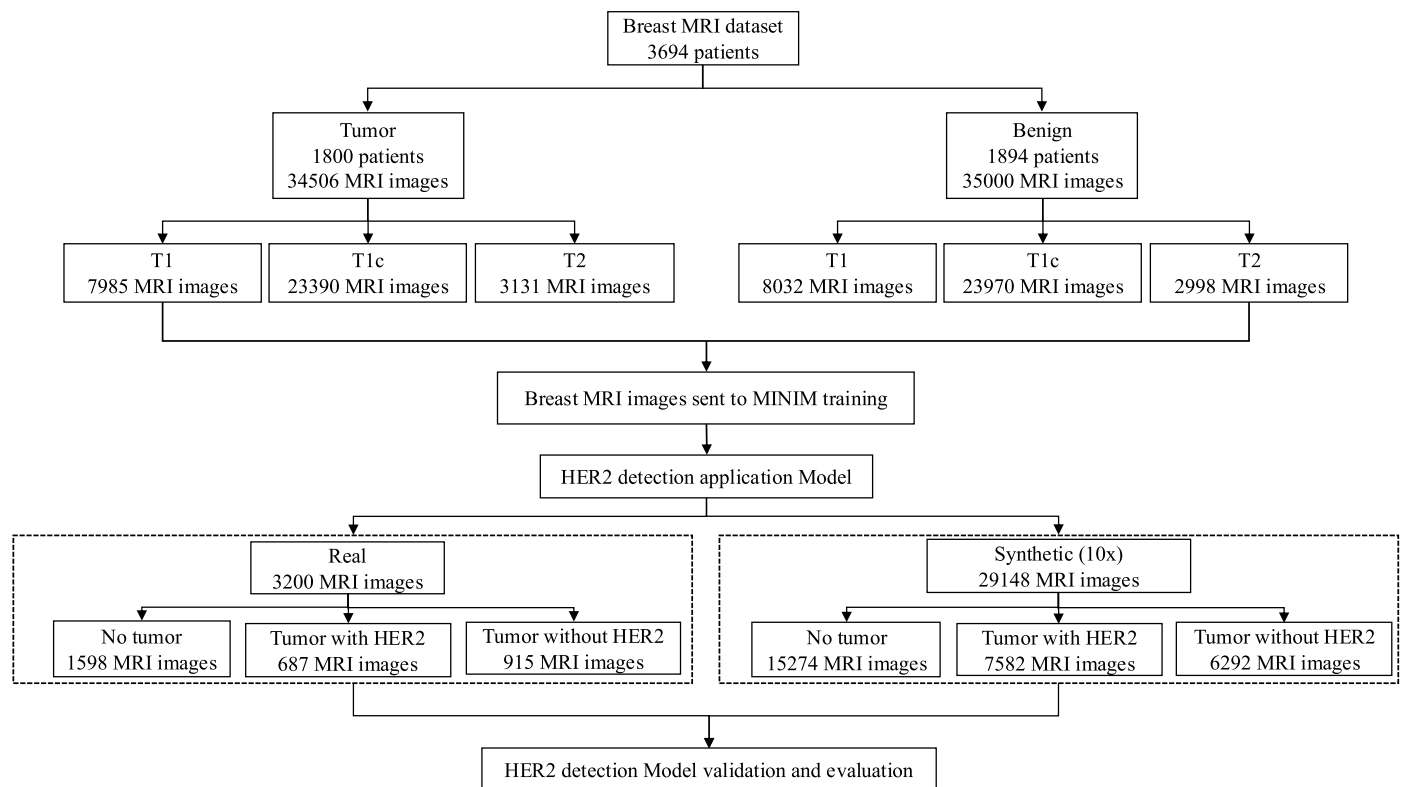


Extended Data Fig. 4 | Synthetic images assist in ophthalmological diagnostic. For diagnostic labels with lower classification performance, MINIM can selectively generate synthetic OCT images and integrate them into the training data, thereby further refining the diagnostic model's accuracy.



Extended Data Fig. 5 | A typical cases of common diseases and report generation. a, Examples of atypical cases of common diseases including retinal vein occlusions (RVOs), diabetic retinopathy (DR), and choroidal neovascularization (CNV). **b**, When compared to reports generated by selected

report generation models and ophthalmologists, the models can identify more comprehensive and accurate information. e, Incorporating OCT image data synthesized by MINIM into the self-supervised learning of OCT image models can enhance the representational capabilities of pre-trained models.



Extended Data Fig. 6 | Breast MRI dataset and analysis flowchart. Breast MR images were split into Tumor and Benign, each of which has three modalities, that is, T1, T1c and T2. These images were sent to MINIM for generative model training and downstream HER2 mutation detection.

Extended Data Table 1 | Data characteristics of the cohorts used in this study

Data characteristics of the OCT cohort				Data characteristics of the fundus cohort									Data characteristics of the lung cancer CT cohort			
	Training Set	Validation Set		Total	RD	AS	GL	ON	DR	ME	AMD	IAOD		EGFR wild type	EGFR-mutation sensitive type	EGFR-mutation resistant type
Number of participants	11,438	2287	Number of participants	27722	1824	2457	3173	287	1644	553	226	206	Number of participants	1750	1930	65
Female (%)	47.6%	50.3%	Female (%)	51.57	62.06	46.81	51.47	50.52	50.49	49.37	39.82	50.00	Female (%)	67.43%	45.03%	49.23%
Age (yr.)	47.09 ± 17.31	46.28 ± 20.27	Age (yr.)	44.7± 22.8	36.88 ± 19.16	18.24 ± 17.28	53.82 ± 19.01	47.87 ± 16.12	60.19 ± 10.75	58.73 ± 14.25	69.44 ± 9.74	36.19 ± 21.11	Age (yr.)	58.4(±11.28)	58.72(±10.36)	56.94(±10.72)
Report Length			Hypertension	457	6	5	29	8	39	9	4	0	Common Mutation	1750(100%)	1829(94.78%)	-
Characters	162.9	179.1	Coronary	4	0	0	1	0	1	0	0	0	Rare Mutation	-	101(5.22%)	-
CLIP Tokens	31.21	35.48	Diabetes	435	1	4	22	2	120	12	2	0	20-INS	-	-	63(96.92%)

RD: Refractive Disorders; AS: Amblyopia and Strabismus; GL (Glaucoma), ON (Optic Neuropathy), DR: Diabetic Retinopathy); ME: Macular Edema; AMD: Age-related Macular Degeneration; IAOD: Inherited and Autoimmune Ocular Disease

Extended Data Table 2 | Ablation study on quantitative assessment of the synthetic images using metrics of fidelity and diversity

	Model	Training Steps	FID	IS	MS-SSIM
Baseline					
	Stable Diffusion (w/o fine-tuning)	-	129.14	2.01	0.55
Training Steps					
	U-Net+CLIP	1k	77.29	2.77	0.48
	U-Net+CLIP	5k	70.88	3.80	0.46
	U-Net+CLIP	10k	65.61	4.82	0.35
Initialization					
	Random U-Net	20k	64.39	2.92	0.26
	Random CLIP	20k	61.79	3.14	0.30
	Random U-Net + CLIP	20k	68.55	2.65	0.45
Resolution					
	Low Resolution	20k	69.43	3.96	0.31
Full Model					
	U-Net + CLIP	20k	57.91	5.89	0.18

The left column indicates different settings of our model. Smaller FID and IS indicate higher fidelity to the original image distribution. Smaller MS-SSIM indicates higher intra-prompt image generation diversity.

Extended Data Table 3 | Ablation study quantitative assessment of the synthetic images in a multi-classification task

	Model	Training Steps	Accuracy				
			CNV	DME	DRUSEN	NORMAL	Macro-Average
Real OCT	-	-	0.97	0.98	0.98	0.98	0.98
	Baseline						
Training Steps	Stable Diffusion (w/o fine-tuning)	-	0.48	0.54	0.46	0.57	0.51
	U-Net + CLIP	1k	0.62	0.68	0.64	0.74	0.67
	U-Net + CLIP	5k	0.80	0.76	0.77	0.82	0.79
	U-Net + CLIP	10k	0.82	0.84	0.88	0.91	0.86
Initialization	Random U-Net	20k	0.88	0.87	0.87	0.94	0.89
	Random CLIP	20k	0.87	0.89	0.89	0.93	0.90
	Random U-Net + CLIP	20k	0.84	0.88	0.82	0.90	0.86
	Resolution						
Full Model	Low Resolution	20k	0.78	0.78	0.82	0.92	0.83
	U-Net + CLIP	20k	0.91	0.93	0.93	0.97	0.94

Multi-label classification performance on the real OCT images and 5,000 synthetic OCT images with different settings is reported.

Extended Data Table 4 | Multi-class diagnosis performance using real OCT images and mixed data of real and synthetic OCT images

Diagnosis name	Diagnosis accuracy (Real.)	Diagnosis accuracy (Real. + Syn.)
Normal	0.96	0.98
Posterior vitreous detachment	0.92	0.97
Epiretinal membrane, before surgery	0.91	0.97
Epiretinal membrane, post-surgery	0.82	0.96
Macular hole, post-surgery	0.55	0.92
Choroidal neovascularization	0.89	0.96
Diabetic macular edema	0.90	0.97
Polypoidal choroidal vasculopathy	0.91	0.97
Endophthalmitis	0.66	0.88
Central serious retinopathy	0.91	0.93
Myopic choroidal neovascularization	0.88	0.96
Choroidal and retinal degeneration	0.67	0.93
Myopia degeneration	0.88	0.98
High myopia chorioretinopathy	0.92	0.98
Drusen	0.92	0.98
Dry age-related macular degeneration, geographic atrophy	0.82	0.91
Wet age-related macular degeneration	0.78	0.93
Branch retinal vein occlusion	0.75	0.89
Central retinal vein occlusion	0.79	0.93
Retinal vasculitis	0.72	0.89
Cystoid macular edema	0.82	0.93
Diabetic retinopathy	0.83	0.90
Uveitis	0.89	0.95
Vitreous hemorrhage	0.90	0.97
Retinal detachment	0.62	0.87

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Software and code

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Data collection	No special software or code was used to collect the data.
Data analysis	We used Pytorch for all data analysis. Custom code was at https://github.com/WithStomach/MINIM .

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We used "sex" in our manuscript because we considered the biological attributes.

Reporting on race, ethnicity, or other socially relevant groupings

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Population characteristics

We collected OCT data and reports from multiple medical institutes. We combined image-report pairs from China Eye Image Investigation and randomly split the data based on patient identifications to build the training and validation sets (11,438 data pairs). We used data from the Eye Hospital of Wenzhou Medical University for external validation (2,287 data pairs). In addition to OCT images with reports from public datasets, we assembled a dataset comprising 13,725 paired image-text OCT samples.

The retinal fundus images were collected from the study cohorts of the China Bioage Investigation Consortium, including the northern China cohort which is from the China suboptimal health cohort study in Tangshan City, Hebei Province, China, and the southern China cohort which is from the Nanfang Hospital in Guangzhou, Guangdong Province, Zhuhai People's Hospital. A total of 59,420 image-text pairs consented to participate in our study.

The chest CT and X-ray images were collected from West China Hospital and Guangzhou Medical University. For each patient with his/her medical record, we selected 10 CT images and 10 X-ray images, resulting in 74,900 paired data for 3,745 patients.

The brain MRI dataset was collected from multiple centers, including 1,960 patients from 16 hospitals in China. These patients collectively contributed a total of 30,800 tumor-annotated images, including 13,859 T1-channel images and 16,941 T2-channel images.

The breast tumor imaging dataset was obtained from the Guangzhou National Laboratory and Sun Yat-sen University Memorial Hospital, encompassing 3,694 patients diagnosed with breast MRI images. These patients provided a total of 34,506 annotated tumor images and 35,000 normal breast images.

Recruitment

All participants provided written informed consent before enrollment.

Ethics oversight

The study was performed in compliance with the tenets of the Declaration of Helsinki and was approved by Institutional Review Board/Ethics Committee of the participated hospitals.

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Sample size

After passing the quality-control step, we used nearly 210 thousands of image-report pairs to train and test our model.

Data exclusions

No data were excluded after passing the initial quality-control step.

Replication

Data replicates were applicable to different splits of the testing sets.

Randomization

Samples were randomly allocated to the training, tuning and testing sets.

Blinding

All data were de-identified to remove any patient-related information.

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